

2020 AMC 10A Problem 10 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 10. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 10

Problem:

Seven cubes, whose volumes are 1, 8, 27, 64, 125, 216, and 343 cubic units, are stacked vertically to form a tower in which the volumes of the cubes decrease from bottom to top. Except for the bottom cube, the bottom face of each cube lies completely on top of the cube below it. What is the total surface area of the tower (including the bottom) in square units?

Answer Choices:

- A. 644
- B. 658
- C. 664
- D. 720
- E. 749

Solution:

The volume of each cube follows the pattern of n^3 , for n is between 1 and 7. We see that the total surface area can be comprised of three parts: the sides of the cubes, the tops of the cubes, and the bottom of the $7 \times 7 \times 7$ cube (which is just $7 \times 7 = 49$). The sides areas can be measured as the sum $4 \sum_{n=1}^7 n^2$, giving us 560. Structurally, if we examine the tower from the top, we see that it really just forms a 7×7 square of area 49. Therefore, we can say that the total surface area is

$$560 + 49 + 49 = \boxed{(B) 658}.$$

Alternatively, for the area of the tops, we could have found the sum

$$\sum_{n=2}^7 ((n)^2 - (n-1)^2)$$

giving us 49 as well.

Note: The area on top and bottom are 49 because the largest area is 49, and the other cubes are "inscribed" in it.

OR

It can quickly be seen that the side lengths of the cubes are the integers from 1 to 7 inclusive.

First, we will calculate the total surface area of the cubes, ignoring overlap. This value is

$$6(1^2 + 2^2 + \cdots + 7^2) = 6 \sum_{n=1}^7 n^2 = 6 \left(\frac{7(7+1)(2 \cdot 7 + 1)}{6} \right) = 7 \cdot 8 \cdot 15 = 840$$

Then, we need to subtract out the overlapped parts of the cubes. Between each consecutive pair of cubes, one of the smaller cube's faces is completely covered, along with an equal area of one of the larger cube's faces. The total area of the overlapped parts of the cubes is thus equal to

$$2 \sum_{n=1}^6 n^2 = 182$$

Subtracting the overlapped surface area from the total surface area, we get

$$840 - 182 = \boxed{(B) 658}.$$

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